

```
In[1]:= Clear["Global`*"];
```

0. Readme

1. $V_{\lambda}^S, V_{\phi}^S, V_r^S, V_{\lambda\lambda}^S, V_{\lambda\phi}^S, V_{\lambda r}^S, V_{\phi\phi}^S, V_{\phi r}^S$, and V_{rr}^S in Table B1 of this paper are represented by SphericalV λ , SphericalV ϕ , SphericalVr, SphericalV $\lambda\lambda$, SphericalV $\lambda\phi$, SphericalV λr , SphericalV $\phi\phi$, SphericalV ϕr , and SphericalVrr.

2. $g_{\lambda}, g_{\phi}, g_r, T_{\lambda\lambda}, T_{\lambda\phi}, T_{\lambda r}, T_{\phi\phi}, T_{\phi r}, T_{r\lambda}, T_{r\phi}$, and T_{rr} in Eqs. (30--32), Tables 1 - 2 of Roussel et al. (2015) are represented by R2015g λ , R2015g ϕ , R2015gr, R2015T $\lambda\lambda$, R2015T $\lambda\phi$, R2015T λr , R2015T $\phi\lambda$, R2015T $\phi\phi$, R2015T ϕr , R2015Tr λ , R2015Tr ϕ , and R2015Trr.

3. The symbols in Roussel et al. (2015) are represented as follows:

- $(X_P, Y_P, Z_P) \rightarrow (x, y, z)$
- $(X_S, Y_S, Z_S) \rightarrow (x', y', z')$, which are represented by (x3, y3, z3)
- $(\Delta X_{PS}, \Delta Y_{PS}, \Delta Z_{PS}) \rightarrow (\Delta x, \Delta y, \Delta z)$
- $(\lambda_P, \theta_P, r_P) \rightarrow (\lambda, \phi, r)$
- $PS \rightarrow l$

4. The integral kernels are considered and the complete expressions should be presented as

$G\rho \int_{\lambda_1}^{\lambda_2} \int_{\phi_1}^{\phi_2} \int_{u_1}^{u_2} \kappa(\text{Kernel}) d\lambda' d\phi' du'$ (i.e. $\int_{\lambda_1}^{\lambda_2} \int_{\phi_1}^{\phi_2} \int_{u_1}^{u_2} G\rho_S(\text{Kernel}) d\Omega_S$, where $d\Omega_S =$

$N_S^3 \cos \phi_S (1 - e^2) u^2 d\lambda d\phi du$ is represented by $\kappa du' d\phi' d\lambda' = N^3 \cos \phi' (1 - e^2) u'^2 du' d\phi' d\lambda'$.

1. GV and GGT expressions in spherical integral kernels from Table B1 of this paper

```
In[2]:= A = x3 * Sin[λ] - y3 * Cos[λ];
        B = x3 * Cos[λ] + y3 * Sin[λ];
```

```
In[4]:= l = Sqrt[r^2 + x3^2 + y3^2 + z3^2 - 2 * r * (z3 * Sin[φ] + B * Cos[φ])];
```

```
In[5]:= SphericalVλ = -A / l^3;
```

```
In[6]:= SphericalVφ = (z3 * Cos[φ] - B * Sin[φ]) / l^3;
```

```
In[7]:= SphericalVr = -(r - z3 * Sin[φ] - B * Cos[φ]) / l^3;
```

```
In[8]:= SphericalVλλ = (-2 * r^2 + x3^2 + y3^2 - 2 * z3^2 - 3 * (x3^2 - y3^2) * Cos[2 * λ] +
                    4 * r * z3 * Sin[φ] + 4 * r * B * Cos[φ] - 6 * x3 * y3 * Sin[2 * λ]) / (2 * l^5);
```

```
In[9]:= SphericalVλφ = -3 * A * Cos[φ] * (z3 - B * Tan[φ]) / l^5;
```

```
In[10]:= SphericalVλr = 3 * A * (r - z3 * Sin[φ] - B * Cos[φ]) / l^5;
```

```
In[11]:= SphericalVφφ = (-r^2 - x3^2 - y3^2 - z3^2 + 3 * z3^2 * (Cos[φ])^2 + 2 * r * z3 * Sin[φ] +
                    2 * B * Cos[φ] * (r - 3 * z3 * Sin[φ]) + 3 * B^2 * (Sin[φ])^2) / l^5;
```

```
In[12]:= SphericalVφr = -3 * (r - z3 * Sin[φ] - B * Cos[φ]) * (z3 * Cos[φ] - B * Sin[φ]) / l^5;
```

```
In[13]:= SphericalVrr = (2 * r^2 - x3^2 - y3^2 - z3^2 -
                    (4 * r - 3 * z3 * Sin[φ] - 3 * B * Cos[φ]) * (z3 * Sin[φ] + B * Cos[φ])) / l^5;
```

2. GV and GGT expressions in spherical integral kernels from Eqs. (30-32), Tables 1 - 2 of Roussel et al. (2015)

$$\begin{aligned} \text{In[14]:= } x &= r * \text{Cos}[\phi] * \text{Cos}[\lambda]; \\ y &= r * \text{Cos}[\phi] * \text{Sin}[\lambda]; \\ z &= r * \text{Sin}[\phi]; \end{aligned}$$

$$\begin{aligned} \text{In[17]:= } \Delta x &= x - x3; \\ \Delta y &= y - y3; \\ \Delta z &= z - z3; \end{aligned}$$

$$\text{In[20]:= } l = \text{FullSimplify}\left[\sqrt{\Delta x^2 + \Delta y^2 + \Delta z^2}\right];$$

$$\text{In[21]:= } R2015g\lambda = \frac{x * y3 - y * x3}{r * \text{Cos}[\phi] * l^3}; (*\text{Eq. 30 of Roussel et al. (2015)*})$$

$$\begin{aligned} \text{In[22]:= } R2015g\phi &= \frac{z * (\text{Cos}[\lambda] * \Delta x + \text{Sin}[\lambda] * \Delta y) - r * \text{Cos}[\phi] * \Delta z}{r * l^3}; \\ &(*\text{Eq. 31 of Roussel et al. (2015)*}) \end{aligned}$$

$$\text{In[23]:= } R2015gr = -\frac{x * \Delta x + y * \Delta y + z * \Delta z}{r * l^3}; (*\text{Eq. 32 of Roussel et al. (2015)*})$$

$$\begin{aligned} \text{In[24]:= } g\lambda\lambda &= \frac{3 * (y * x3 - x * y3)^2}{r * \text{Cos}[\phi] * l^5} + \frac{x * \Delta x + y * \Delta y - r^2 * (\text{Cos}[\phi])^2}{r * \text{Cos}[\phi] * l^3}; \\ &(*\frac{\partial(dg\lambda)}{\partial\lambda\rho} \text{ in Table 2 of Roussel et al. (2015)*}) \end{aligned}$$

$$\begin{aligned} \text{In[25]:= } g\lambda\phi &= \frac{z * \text{Cos}[\phi] * (\text{Cos}[\lambda] * \Delta y - \text{Sin}[\lambda] * \Delta x) + \text{Sin}[\phi] * (y3 * x - x3 * y)}{r * \text{Cos}[\phi] * l^3} + \\ &\frac{3 * (x * y3 - y * x3) * (z * (\text{Cos}[\lambda] * \Delta x + \text{Sin}[\lambda] * \Delta y) - r * \text{Cos}[\phi] * \Delta z)}{r * \text{Cos}[\phi] * l^5}; \\ &(*\frac{\partial(dg\lambda)}{\partial\phi\rho} \text{ is revised from Table 2 of Roussel et al. (2015)*}) \end{aligned}$$

$$\begin{aligned} \text{In[26]:= } g\lambda r &= \frac{3 * (y * x3 - x * y3) * (x * \Delta x + y * \Delta y + z * \Delta z)}{r^2 * \text{Cos}[\phi] * l^5}; \\ &(*\frac{\partial(dg\lambda)}{\partial r\rho} \text{ in Table 2 of Roussel et al. (2015)*}) \end{aligned}$$

$$\begin{aligned} \text{In[27]:= } g\phi\lambda &= \frac{3 * (x * y3 - y * x3) * (z * (\text{Cos}[\lambda] * \Delta x + \text{Sin}[\lambda] * \Delta y) - r * \text{Cos}[\phi] * \Delta z)}{r * l^5} + \\ &\frac{z * (\text{Cos}[\phi])^2 * (\text{Cos}[\lambda] * \Delta y - \text{Sin}[\lambda] * \Delta x) - \text{Sin}[\phi] * (1 - \text{Cos}[\phi]) * (y3 * x - x3 * y)}{r * \text{Cos}[\phi] * l^3}; \\ &(*\frac{\partial(dg\phi)}{\partial\lambda\rho} \text{ is revised from Table 2 of Roussel et al. (2015)*}) \end{aligned}$$

$$\begin{aligned} \text{In[28]:= } g\phi\phi &= \frac{3 * (z * (\text{Cos}[\lambda] * \Delta x + \text{Sin}[\lambda] * \Delta y) - r * \text{Cos}[\phi] * \Delta z)^2}{r * l^5} + \frac{x * \Delta x + y * \Delta y + z * \Delta z - r^2}{r * l^3}; \\ &(*\frac{\partial(dg\phi)}{\partial\phi\rho} \text{ in Table 2 of Roussel et al. (2015)*}) \end{aligned}$$

```

In[29]:= gφr = 
$$\frac{3 * (r * \text{Cos}[\phi] * \Delta z - z * (\text{Sin}[\lambda] * \Delta y + \text{Cos}[\lambda] * \Delta x)) * (x * \Delta x + y * \Delta y + z * \Delta z)}{r^2 * l^5};$$

(*  $\frac{\partial(dg\phi)}{\partial r p}$  in Table 2 of Roussel et al. (2015) *)

In[30]:= grλ = 
$$\frac{3 * (y * x^3 - x * y^3) * (x * \Delta x + y * \Delta y + z * \Delta z)}{r * l^5} - \frac{y * x^3 - x * y^3}{r * l^3};$$

(*  $\frac{\partial(dgr)}{\partial \lambda p}$  in Table 2 of Roussel et al. (2015) *)

In[31]:= grφ = 
$$\frac{(r * \text{Cos}[\phi] * \Delta z - z * (\text{Cos}[\lambda] * \Delta x + \text{Sin}[\lambda] * \Delta y)) * (3 * (x * \Delta x + y * \Delta y + z * \Delta z) - l^2)}{r * l^5};$$

(*  $\frac{\partial(dgr)}{\partial \phi p}$  in Table 2 of Roussel et al. (2015) *)

In[32]:= grr = 
$$\frac{3 * (x * \Delta x + y * \Delta y + z * \Delta z)^2}{r^2 * l^5} - \frac{1}{l^3};$$
 (*  $\frac{\partial(dgr)}{\partial r p}$  in Table 2 of Roussel et al. (2015) *)

In[33]:= R2015Tλλ = FullSimplify[
$$\frac{1}{r * \text{Cos}[\phi]} * g\lambda\lambda + \frac{1}{r} * R2015gr - \frac{\text{Sin}[\phi]}{r * \text{Cos}[\phi]} * R2015g\phi$$
];
(*dTλλ in Table 1 of Roussel et al. (2015) *)

In[34]:= R2015Tλφ = FullSimplify[
$$\frac{1}{r} * g\lambda\phi$$
]; (*dTλφ in Table 1 of Roussel et al. (2015) *)

In[35]:= R2015Tλr = FullSimplify[gλr]; (*dTλr in Table 1 of Roussel et al. (2015) *)

In[36]:= R2015Tφλ = FullSimplify[
$$\frac{1}{r * \text{Cos}[\phi]} * g\phi\lambda + \frac{\text{Sin}[\phi]}{r * \text{Cos}[\phi]} * R2015g\lambda$$
];
(*dTφλ in Table 1 of Roussel et al. (2015) *)

In[37]:= R2015Tφφ = FullSimplify[
$$\frac{1}{r} * g\phi\phi + \frac{1}{r} * R2015gr$$
];
(*dTφφ in Table 1 of Roussel et al. (2015) *)

In[38]:= R2015Tφr = FullSimplify[gφr]; (*dTφr in Table 1 of Roussel et al. (2015) *)

In[39]:= R2015Trλ = FullSimplify[
$$\frac{1}{r * \text{Cos}[\phi]} * gr\lambda - \frac{1}{r} * R2015g\lambda$$
];
(*dTTrλ in Table 1 of Roussel et al. (2015) *)

In[40]:= R2015Trφ = FullSimplify[
$$\frac{1}{r} * gr\phi - \frac{1}{r} * R2015g\phi$$
];
(*dTTrφ in Table 1 of Roussel et al. (2015) *)

In[41]:= R2015Trr = FullSimplify[grr]; (*dTrr in Table 1 of Roussel et al. (2015) *)

In[42]:= FullSimplify[R2015Tλφ - R2015Tφλ]
Out[42]=
0

In[43]:= FullSimplify[R2015Tλr - R2015Trλ]
Out[43]=
0

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```
In[44]:= FullSimplify[R2015T $\phi$ r - R2015Tr $\phi$ ]
Out[44]=
0
```

```
In[45]:= Laplace1 = FullSimplify[R2015T $\lambda\lambda$  + R2015T $\phi\phi$  + R2015Trr]
Out[45]=
0
```

3. Differences of GV and GGT expressions in spherical integral kernels between Table B1 of this paper and Eqs. (30-32), Tables 1 - 2 of Roussel et al. (2015)

```
In[46]:= FullSimplify[SphericalV $\lambda$  - R2015g $\lambda$ ]
Out[46]=
0
```

```
In[47]:= FullSimplify[SphericalV $\phi$  - R2015g $\phi$ ]
Out[47]=
0
```

```
In[48]:= FullSimplify[SphericalVr - R2015gr]
Out[48]=
0
```

```
In[49]:= FullSimplify[SphericalV $\lambda\lambda$  - R2015T $\lambda\lambda$ ]
Out[49]=
0
```

```
In[50]:= FullSimplify[SphericalV $\lambda\phi$  - R2015T $\lambda\phi$ ]
Out[50]=
0
```

```
In[51]:= FullSimplify[SphericalV $\lambda$ r - R2015T $\lambda$ r]
Out[51]=
0
```

```
In[52]:= FullSimplify[SphericalV $\phi\phi$  - R2015T $\phi\phi$ ]
Out[52]=
0
```

```
In[53]:= FullSimplify[SphericalV $\phi$ r - R2015T $\phi$ r]
Out[53]=
0
```

```
In[54]:= FullSimplify[SphericalVrr - R2015Trr]
Out[54]=
0
```

```
In[55]:= NotebookSave[EvaluationNotebook[]];
```